**Capstone FA26 through SP27**

| x' indicates cross sectic | Stakeholder | Stakeholder Email | Team Member | Student Email | Project Title & Description | |
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| **Team1x** | **Mohammad Taha 1** | **mtaha3@student.gsu.edu** | Alireza Firouzi<br>Prosper Olusegun-Joseph<br>Bryant Awoh<br>Keemani Pedro<br>Femi Oduba | afirouzi1@student.gsu.edu<br>polusegunjoseph2@student.gsu.edu<br>bawoh1@student.gsu.edu<br>kpedro1@student.gsu.edu<br>ooduba1@student.gsu.edu | **Study Buddy:** app that students can use to find other people to work with in a class.<br>Students can sign up as individuals or groups and list classes they are in and their previous software<br>Projects, suggest pairs and share contact information when confirmed, chat room for initial conversations with group members. Possible to use GenAI and upload resume, transcript to match. | |
| **Team2x** | **Dr. Johnson 1** | **wjohnson6@gsu.edu** | Abraham Abraham<br>Lauriane Houndjahoue<br>Z Hilden-Minton<br>Mamadou Diallo<br>Remlla Ali | kabraham6@student.gsu.edu<br>lhoundjahoue1@student.gsu.edu<br>zhildenminton1@student.gsu.edu<br>mdiallo36@student.gsu.edu<br>rali24@student.gsu.edu | **An Online Repository for Capstone Project Selection and Organization**: App keeps numerous Capstone projects and provides a workspace for team formation and collaboration based on project topic.<br>Hosting project ideas for students to pick from to work on a project instead of team creation first.<br>A dashboard for course students looking for teammates based upon common interests in a project. | |
| **Team3 custom** | **Dr. Johnson 2** | **wjohnson6@gsu.edu** | Chukwudalu Nwosu<br>Dignity Okorie<br>Femi Ijimayowa<br>Jim-nelson Shillingford | cnwosu9@student.gsu.edu<br>dokorie1@student.gsu.edu<br>fijimayowa1@student.gsu.edu<br>jshillingford1@student.gsu.edu | (custom) **Car battery mgmt. system.** The purpose of our project is to replicate the Tesla car battery management system by providing a thermal model for 3 devices (Radio, Lightbulb, Fan). Each device can be turned on and off through our website and is powered by a Raspberry Pi microcontroller; users can also see each device's battery percentage and a detailed explanation of which device uses the most power.<br>The significance of this project is to show the interaction between hardware and software and how it is used it every-day devices such as a car.<br>Project Goals<br>1. To develop embedded skills that are transferable to other sectors in computer science<br>2. To become more efficient in backend and front end frameworks<br>3. Learn to collaborate with other engineers and be able to proficient in version control software like GitHub. | |
| **Team4** | **Mohammad Taha 2** | **mtaha3@student.gsu.edu** | Alexa Veloz<br>Deepak Govindarajan<br>Dennis Golenkov<br>Ryan Semancik<br>Trajuan Smith | aveloz1@student.gsu.edu<br>dg1@student.gsu.edu<br>dgolenkov1@student.gsu.edu<br>rsemancik3@student.gsu.edu<br>tsmith413@student.gsu.edu | **Mobile App Check-In System for University Tutoring Services**. This project aims to develop a mobile app check-in system for undergraduate students using a university tutoring service. The app will utilize dynamic QR codes to streamline the check-in process, making it more efficient and convenient for both students and tutors | |
| **Team5** | **Dr. Johnson 3** | **wjohnson6@gsu.edu** | Jaimere Heath<br>Kevin Pham<br>Keya Yonas<br>Stefanny Do<br>Toan Nguyen | jheath12@student.gsu.edu<br>kpham41@student.gsu.edu<br>kyonas1@student.gsu.edu<br>sdo7@student.gsu.edu<br>tnguyen665@student.gsu.edu | **Research Portal for Undergraduate Students**<br>Objective: System to manage matching UG student with research in CS<br>Constraints: Use AI to scan research documents, student CV, student records, student GIT repository, LinkedIn, etc. Compile to create student matching values with research requirements. Ux would be instructor and student. Could also include statement of purpose, recommendation letters, etc. | |
| **Team6** | **Dr. Tamhankar** | **ptamhankar@gsu.edu** | Denis Joseph<br>Mikhail Tsegaye<br>Tauha Hysaw<br>Taizo Rashid<br>Vince Paul | djoseph22@student.gsu.edu<br>mtsegaye1@student.gsu.edu<br>thysaw1@student.gsu.edu<br>trashid5@student.gsu.edu<br>vpaul5@student.gsu.edu | **Intelligent Assignment Similarity and Derivation Analysis System (IAS-DAS) * Faculty Stakeholder**<br>Objective: Design and develop an intelligent decision-support system for analyzing student assignments to identify potentially similar or derived submissions. The system will explore techniques that go beyond traditional text matching by examining different forms of similarity and potential AI-assisted transformations. The framework is intended to be generalized to different types of assignments, with programming assignments serving as the initial implementation and evaluation domain. | |

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| **Team7** | **Anil Tiwari**<br>**1** | **atiwari12@student.gsu.edu** | Amanda Peart<br>Kayla Peters<br>Diarra Sarr<br>Sabrina Beshir | apeart5@student.gsu.edu<br>kpeters19@student.gsu.edu<br>msarr4@student.gsu.edu<br>sbeshir2@student.gsu.edu | **Tiny Task**s<br>**Objective**: Schedule management with accessibility focused for ADHD people.<br>**Constraints**:<br>•Given that almost 30% of undergraduate students are taking some type of ADHD medication, managing daily/weekly tasks for school and life is overwhelming<br>•Simple mobile app that allows student to see only daily task list or even ½ day<br>•Shall have an interface to Google calendar<br>•Shall use AI to determine if task can be broken down into smaller ones | |
| **Team8x** | **Anil Tiwari**<br>**2** | **atiwari12@student.gsu.edu** | Emilia Bravo<br>Hafeez Imran<br>Raina Won<br>Jerry Hill II<br>Pavon Gurung | ebravodedo1@student.gsu.edu<br>himran2@student.gsu.edu<br>rwon1@student.gsu.edu<br>jhill149@student.gsu.edu<br>ygurung2@student.gsu.edu | **AI CS Program Information Chatbot**<br>**Objective**: Build an AI chatbot that can be used in the department for answering FAQs and student inquiries.<br>**Constraints**:<br>•AI automation of answering CS student inquiries<br>•Can develop it for both graduate and undergraduate program first by training it on the student's handbook. | |
| **Team9x** | **Mohammad Taha**<br>**3** | **mtaha3@student.gsu.edu** | Andrew Dana<br>Eyobed Gebregziabher<br>Raphael Omorose<br>Uyiosa Nehikhuere | adana1@student.gsu.edu<br>egebregziabher1@student.gsu.edu<br>romorose1@student.gsu.edu<br>unehikhuere1@student.gsu.edu | **Face-IT.** A university attendance app using facial scans leverages AI-driven facial recognition to identify and verify students, automatically marking their presence and logging attendance data in a database for accurate record-keeping and reporting. Key benefits include enhanced security, automation of attendance tracking, increased accuracy, and reduced manual effort for educators. | |
| **Team10 custom** | **Dr. Johnson**<br>**4** | **wjohnson6@gsu.edu** | Amaya Rue<br>Crow Veal<br>Long Dinh<br>Rich Nguyen | arue2@student.gsu.edu<br>hveal3@student.gsu.edu<br>ldinh11@student.gsu.edu<br>rnguyen32@student.gsu.edu | **(custom) Small Tasks Saves Time — Goal breakdown**<br>Provide a brief overview of the project, including the problem it aims to solve or the need it addresses.<br>The problem the application seeks to address is the ambiguity regarding how one can achieve a goal. It will give the user the opportunity to break down their goal into more digestible steps and provide a list with what they have decided. | |
| **Team11** | **Anil Tiwari**<br>**3** | **atiwari12@student.gsu.edu** | Hoang-Long Do<br>Huy Nguyen<br>Phuc Nguyen<br>Parth Patel<br>Tin Chak Lam | hdo19@student.gsu.edu<br>hnguyen296@student.gsu.edu<br>pnguyen143@student.gsu.edu<br>ppatel282@student.gsu.edu<br>tlam31@student.gsu.edu | **Food Oasis.** A mobile app that uses GPS mapping for finding purveyors of fresh food items and purchase for pickup (no delivery). Simple interface using GPS to find fresh food purveyors. Users looking for food do not need to register to find food and shall have working GPS to use the system. | |
| **Team12x custom** | **Dr. Mikler**<br>**and Samuel Parisa** | **amikler@gsu.edu**<br>**sparisa1@gsu.edu** | Aisha Yahya<br>Shreya Akula<br>Nameera Afrose<br>Sushmitha Dumpala<br>Vishnavi Abhishikta Bollapalli<br>Ishita Tambekar | ayahya3@student.gsu.edu<br>laxmi@student.gsu.edu<br>nafrose1@student.gsu.edu<br>sdumpala1@student.gsu.edu<br>vbollapalli1@student.gsu.edu<br>itambekar1@student.gsu.edu | **(custom) ASL modules**<br>The system will use a webcam to recognize supported ASL signs and convert them into English text. We plan to build the project in stages, starting with individual signs and gradually working toward sentences, longer responses, and interview communication. Depending on how the system performs, we also want to explore how it could be used with platforms such as Zoom or Microsoft Teams.<br>This project combines computer vision, machine learning, web development, databases, and accessibility to address a real communication problem. | |
| **Team13x custom** | **Dr. Sadasivuni** | **ssadasivuni1@gsu.edu** | Armando Galvan<br>Brendon Huang<br>Laurel Porter<br>Jamal Lyons<br>Karan Narain | agalvan2@student.gsu.edu<br>bhuang6@student.gsu.edu<br>lporter11@student.gsu.edu<br>jlyons20@student.gsu.edu<br>knarain1@student.gsu.edu | **FoundU**<br><br>Losing personal belongings on a college campus can be frustrating, especially when lost-and-found information is scattered across different social media platforms, group chats, and campus offices. FoundU is a centralized lost-and-found application designed specifically for college campuses. It allows students to report, search for, and recover lost or found items in one place rather than relying on multiple platforms or informal methods. Users can create reports that include details such as an item's description, where it was lost or found, and the approximate date and time. The system would then compare reports to identify potential matches and make the recovery process more efficient. | |

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| **Team14x** | **Mohammad Taha**<br>**4** | **mtaha3@student.gsu.edu** | Thien Nguyen<br>Ken Tran<br>Melvin Ibarra Robledo<br>Rishik Poosapati<br>Rishi Shah | cnguyen130@student.gsu.edu<br>ktran92@student.gsu.edu<br>mibarrarobledo2@student.gsu.edu<br>rpoosapati1@student.gsu.edu<br>rshah38@student.gsu.edu | **Tiny Tasks**<br>**Objective:** Schedule management with accessibility focused for ADHD people.<br>**Constraints:**<br>Given that almost 30% of undergraduate students are taking some type of ADHD medication, managing daily/weekly tasks for school and life is overwhelming<br>Simple mobile app that allows student to see only daily task list or even ½ day<br>Shall have an interface to Google calendar<br>Shall use AI to determine if task can be broken down into smaller ones | |
| **Team15 custom** | **Dr. Johnson**<br>**5** | **wjohnson6@gsu.edu** | Hoang Nguyen<br>Issac Gao<br>Julia Wei<br>Samuel Hobbs<br>Tina Tran | hnguyen275@student.gsu.edu<br>igao2@student.gsu.edu<br>jwei9@student.gsu.edu<br>shobbs15@student.gsu.edu<br>ttran225@student.gsu.edu | (custom) Universal Music Player<br>**Introduction**<br>•With Android often lacking a built-in local music player besides basic support in the files app, our group is developing a dedicated music app that can play our downloaded music library with much more fidelity and combine it with songs from music streaming services.<br>•This app will be able to sort and play local files with the same functionality that many music streaming services have. It will be able to build playlists regardless of file structure, and songs can appear on multiple playlists. It will also support music sourced from an external music streaming API the user is registered in.<br>•This project would have us handling databases to track available music files, connecting APIs for external music sourcing, and managing audio I/O to play the music. | |
| **Team16x** | **Mohammad Taha**<br>**5** | **mtaha3@student.gsu.edu** | Cor'layshia Cooper<br>Derick Nguyen<br>Taylor Glascock<br>Neo Avecillas Deleg | ccooper97@student.gsu.edu<br>dnguyen209@student.gsu.edu<br>tglascock1@student.gsu.edu<br>navecillasdeleg1@student.gsu.edu | **Food Oasis**<br>Objective: A mobile app that uses GPS mapping for finding purveyors of fresh food items and purchase for pickup (no delivery).<br><br>Constraints:<br>•Simple interface using GPS to find fresh food purveyors<br>•Users looking for food do not need to register to find food and shall have working GPS to use the system<br>•Users wanting to reserve food will need to register with Food Oasis<br>•Suppliers will need to register, like eBay seller, and input products<br>•GPS connections are focus | |
| **Team17** | **Anil Tiwari**<br>**4** | **atiwari12@student.gsu.edu** | Mohammed Afeefuddin<br>Mohammed Hussain Shareef Shaik<br>Parth Chavan<br>Shuja Mukhtar Syed | mafeefuddin1@student.gsu.edu<br>mshaik3@student.gsu.edu<br>pchavan3@student.gsu.edu<br>smukhtar2@student.gsu.edu | **Face-IT.** A university attendance app using facial scans leverages AI-driven facial recognition to identify and verify students, automatically marking their presence and logging attendance data in a database for accurate record-keeping and reporting. Key benefits include enhanced security, automation of attendance tracking, increased accuracy, and reduced manual effort for educators. | |
| **Team18** | **Anil Tiwari**<br>**5** | **atiwari12@student.gsu.edu** | Carlos Taylor<br>Michael Jordan<br>Malaika Patounas<br>Oluchi Chukwurah<br>Temiloluwa Alajogun | ctaylor150@student.gsu.edu<br>mjordan68@student.gsu.edu<br>mpatounas1@student.gsu.edu<br>ochukwurah1@student.gsu.edu<br>talajogun1@student.gsu.edu | **Study Buddy:** app that students can use to find other people to work with in a class.<br>Students can sign up as individuals or groups and list classes they are in and their previous software<br>Projects, suggest pairs and share contact information when confirmed, chat room for initial conversations with group members. Possible to use GenAI and upload resume, transcript to match. | |

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| **Team19** | **Dr. Venkateswara** | **hvenkateswara@gsu.edu** | Anh Huynh<br>Brooke Hasty<br>Kenny Cao<br>Snigdha Addagarla<br>Seul An Kim | ahuynh24@student.gsu.edu<br>bhasty1@student.gsu.edu<br>kcao9@student.gsu.edu<br>saddagarla1@student.gsu.edu<br>skim363@student.gsu.edu | **Document-Grounded AI Assistant for the CS Program * Faculty Stakeholder**<br>**Objective**: Develop an AI assistant that answers student questions about CS programs, courses, degree requirements, policies, forms, and procedures using an approved collection of department/university documents.<br><br>**Constraints / Features:**<br>•Use a RAG architecture to retrieve information from approved documents.<br>•LLM may interpret questions and formulate responses, but answers must be grounded in retrieved sources.<br>•Every factual answer should provide source citations.<br>•If adequate supporting information cannot be found, the chatbot should decline to answer or direct the student to the appropriate office.<br>•Administrative interface for adding, removing, updating, and re-indexing documents.<br>•Support PDFs, webpages, and structured FAQ content.<br>•Logging and feedback mechanisms for identifying unanswered or incorrectly answered questions.<br>•Evaluate retrieval accuracy, groundedness, citation correctness, response latency, and usability. | |
| **Team20 custom** | **Dr. Johnson 6** | **wjohnson6@gsu.edu** | Andrii Fil<br>Cittlali Romero<br>Phu Ly<br>Samrin Zaman<br>Vihaan Dhaka | afil1@student.gsu.edu<br>cromero12@student.gsu.edu<br>ply8@student.gsu.edu<br>szaman12@student.gsu.edu<br>vdhaka1@student.gsu.edu | (custom) **Panther Carts \| Campus Cart Rental Queue Management**<br>**Introduction**<br>•Panther Carts is a web-and-SMS system that will manage rentals of numbered cart bins during staffed campus-housing move-in sessions. It replaces fragmented sign-up sheets and verbal updates with a coordinated flow for student registration, fair queue placement, pickup, checkout, return, and staff and administrator oversight.<br>•The project addresses a practical resource-allocation problem: demand can exceed the number of available carts, students need timely status updates, and staff must match each physical bin and retain PantherCard to the correct student without losing queue order.<br>•Panther Carts is significant in computer science because it combines user-centered web design, concurrent database transactions, secure access control, real-time operational reporting, and provider-independent messaging. Its central challenge is keeping physical and digital state consistent while several users may act at the same time. | |
| **Team21** | **Anil Tiwari 6** | **atiwari12@student.gsu.edu** | Destine Navarrete<br>Evelyn Hernandez<br>Jihun Kim<br>Yanshi Patel | dnavarrete4@student.gsu.edu<br>ehernandez45@student.gsu.edu<br>jkim529@student.gsu.edu<br>ypatel44@student.gsu.edu | **Commute Helper**<br>Objective: A web app that analyzes common commute methods for GSU students and displays updates on delays and other events affecting transit data, traffic delays, and student parking status.<br>**Constraints**:<br>●App should display data on Marta delays and nearby traffic<br>●App should regularly update/refresh with new information | |
| **Team22** | **Anil Tiwari 7** | **atiwari12@student.gsu.edu** | Wright Christian<br>Mandzyuk David<br>Urum Emmanuel<br>Cortez George | cwright133@student.gsu.edu<br>dmandzyuk1@student.gsu.edu<br>eurum1@student.gsu.edu<br>gcortez4@student.gsu.edu | **Translate-point**<br>Objective: A web enabled application that consumes a PowerPoint (native pptx) file and translates into another language using an AI system.<br>**Constraints**:<br>•Simple graphical interface using a file 'upload' look and feel<br>•Users will be prompted for translate 'from' 'to' language | |

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| **Team23** | **Mohammad Taha 6** | **mtaha3@student.gsu.edu** | Kolagani Joel<br>Villeda Martinez Jair<br>Callen Rocket<br>Higginbotham Xavier | jkolagani1@student.gsu.edu<br>jvilledamartinez1@student.gsu.edu<br>rcallen1@student.gsu.edu<br>xhigginbotham1@student.gsu.edu | **A system to judge poster submissions**<br>**Objective**: A software application, which may include a combination of a mobile app and a central dashboard.<br>**Constraints**:<br>●A mobile app for judges to input scores from a predefined rubric.<br>●The pertinent rubric must be retrieved based on the type of submission being judged – For example, Ph.D. poster, Demo, etc.<br>●A central dashboard that aggregates scores for all submissions and ranks submissions based on the aggregate score. | |
| **Team24** | **Mohammad Taha 7** | **mtaha3@student.gsu.edu** | Antwione Day<br>Bolu Ajayi<br>David Choi<br>Imad Shah | aday28@student.gsu.edu<br>bajayi2@student.gsu.edu<br>dchoi24@student.gsu.edu<br>ishah5@student.gsu.edu | **Mobile App Check-In System for University Tutoring Services**<br>**Objective**: This project aims to develop a mobile app check-in system for undergraduate students using a university tutoring service. The app will utilize dynamic QR codes to streamline the check-in process, making it more efficient and convenient for both students and tutors.<br>**Constraints**:<br>•QR code generation: Tutors will be able to generate unique QR codes for each session.<br>•Student check-in: Students will scan the QR code using the mobile app to check in for their tutoring session.<br>•Real-time attendance data: Attendance data will be automatically recorded and accessible to tutors and administrators in real-time.<br>•Session details: The app will display relevant session details, including tutor information, session time, and subject.<br>•Push notifications: Students and tutors will receive push notifications reminding them of upcoming sessions and check-in times.<br>•Offline functionality: The app will function even without an internet connection, ensuring uninterrupted check-in even in areas with limited connectivity.<br>•A server-side component to manage QR code generation, data storage, and user authentication. | |
| **Team25** | **Dr. Weeks** | **mweeks@gsu.edu** | Josiah Jones<br>Jaden Woody<br>Kamari Darrisaw<br>Musab Adeogun | jjones502@student.gsu.edu<br>jwoody6@student.gsu.edu<br>kdarrisaw1@student.gsu.edu<br>madeogun1@student.gsu.edu | **Raspberry Pi Super Mario * Faculty Stakeholder**<br>**Objective**: control a game character with a Raspberry Pi<br><br>**Constraints**: Games such as Super Mario (https://supermarioplay.com/) can be played by a human with a keyboard, using the arrow keys and the space bar. A Raspberry Pi is a small computer that can do many things, including capturing images/video, running AI programs, and connecting to other devices via Bluetooth (see HID keyboards). Could we have Raspberry Pi play a game like Super Mario by getting input (through a camera focused on another computer's monitor), processing the game state, and sending output to the other computer (in the form of simulated key presses)? That is the goal. | |
| **Team26** | **Dr. James** | **djames56@gsu.edu** | Omer Ohayon<br>Jason Li<br>Ryzin Smith<br>Yash Saini | oohayon1@student.gsu.edu<br>qli26@student.gsu.edu<br>rsmith283@student.gsu.edu<br>ysaini2@student.gsu.edu | **Monitor the supply chain for tariffs and provide an alert if a potential price increase is detected.**<br>**Objective**: The software application must contain alerting user(s) about a potential shortage resulting from supply chain disruption; an example of such a scenario is the recent baby formula shortage.<br>**Constraints**:<br>●A mobile app/web dashboard that can display the current "health" of a supply chain (students may pick one or two essential goods as a case study)<br>●The software application must raise an alert as soon as a potential disruption is detected.<br>●Additionally, the app must also scour inventory from local stores and display listings by price and/or distance from the user. | |